

Lesson 06 Demo 05

Setting Up and Configuring Azure Virtual Machine Scale Sets

Objective: To set up and configure Azure virtual machines for high availability using Virtual Machine Scale Sets (VM Scale Sets)

Tools required: Azure account with administrator access

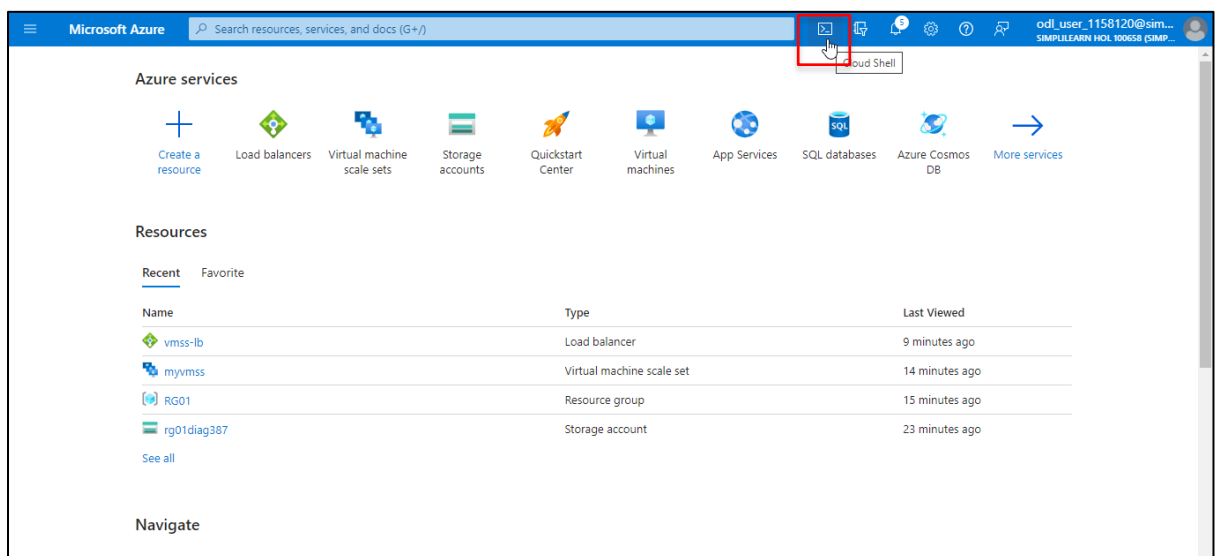
Prerequisites: None

Steps to be followed:

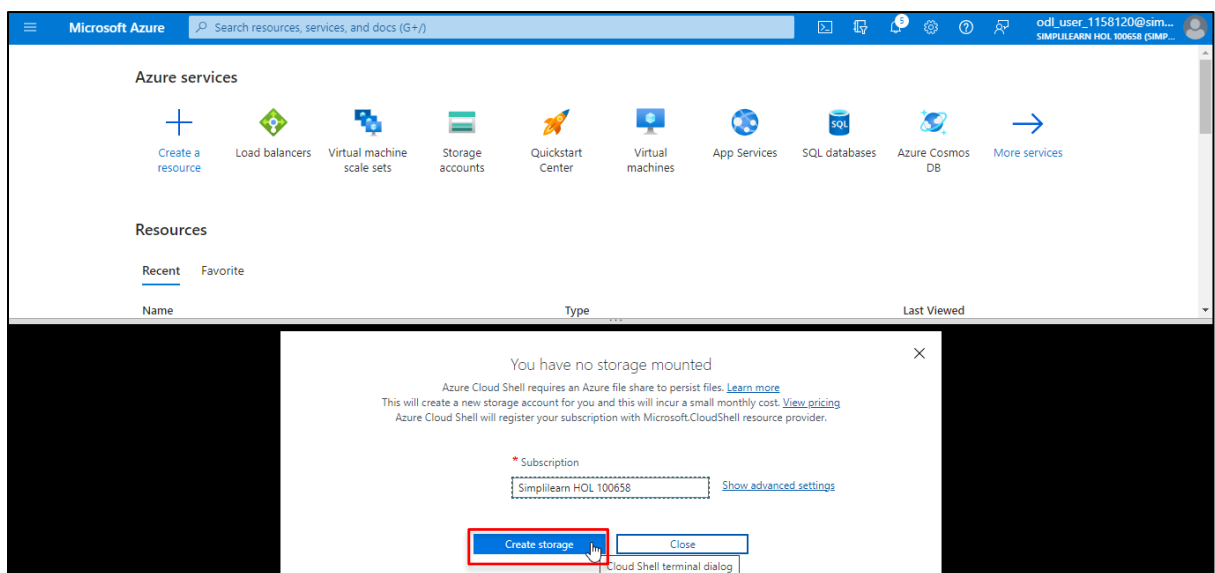
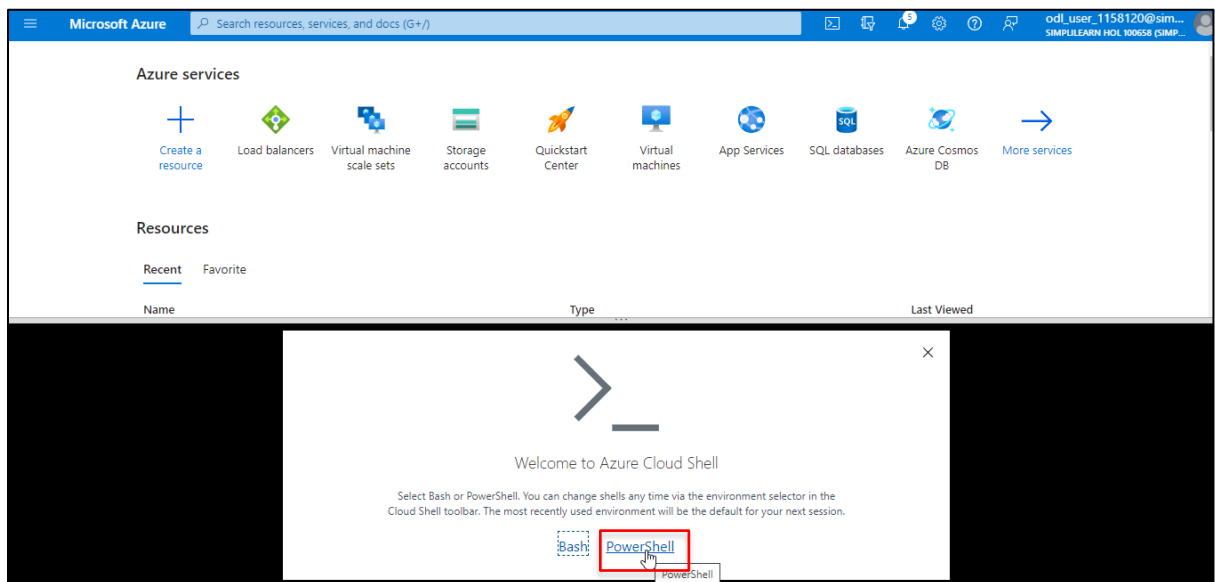
1. Register the Microsoft resource providers
2. Deploy zone-resilient Azure VM Scale Sets
3. Configure Azure VM Scale Sets using virtual machine extensions

Step 1: Register the Microsoft resource providers

1.1 Click on **Cloud Shell** after logging in to the Azure portal



1.2 Select PowerShell and click on Create storage



1.3 Run the following commands to register the Microsoft resource providers:

Register-AzResourceProvider -ProviderNamespace Microsoft.Insights

```
PS /home/odl_user> Register-AzResourceProvider -ProviderNamespace Microsoft.Insights

ProviderNamespace : microsoft.insights
RegistrationState  : Registered
ResourceTypes     : {components, components/query, components/metadata, components/metrics...}
Locations         : {East US, South Central US, North Europe, West Europe...}
```

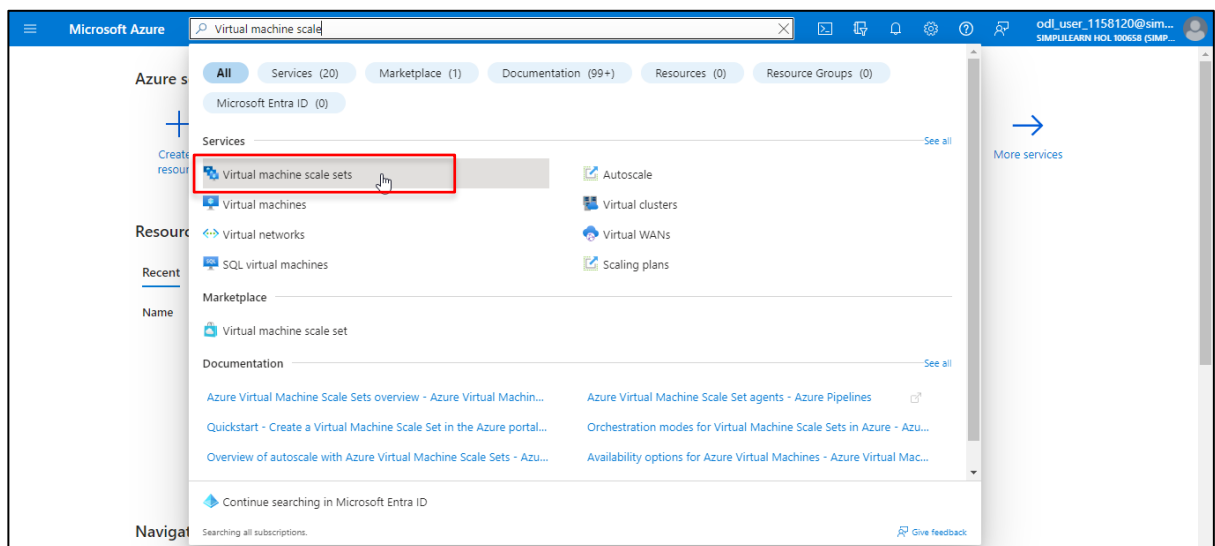
Register-AzResourceProvider -ProviderNamespace Microsoft.AlertsManagement

```
PS /home/odl_user> Register-AzResourceProvider -ProviderNamespace Microsoft.AlertsManagement

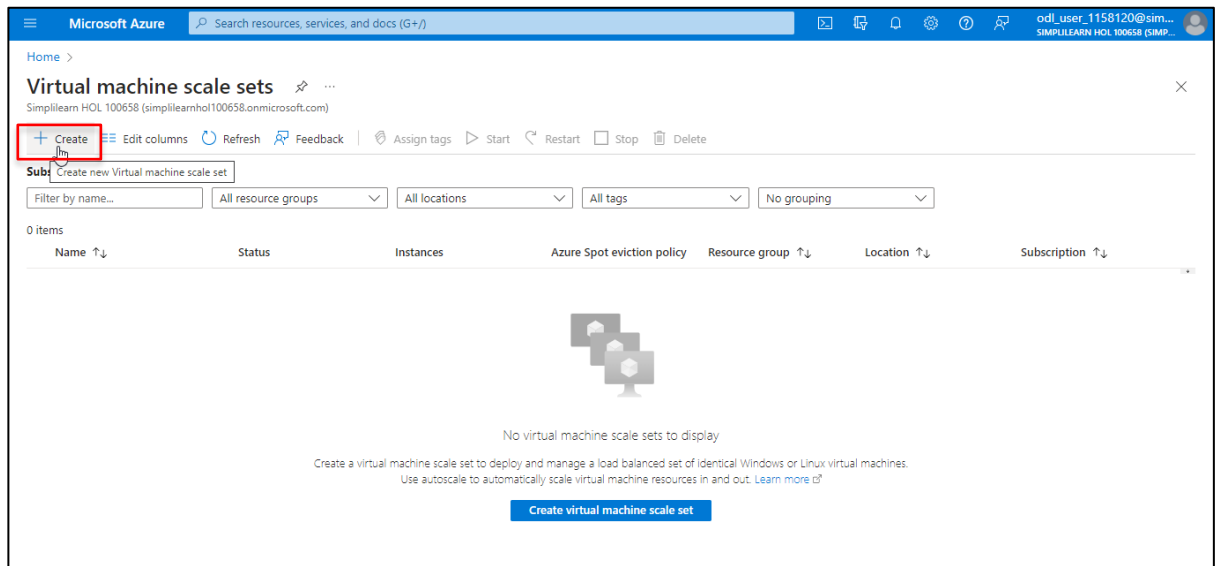
ProviderNamespace : Microsoft.AlertsManagement
RegistrationState  : Registered
ResourceTypes     : {alerts, alertsSummary, smartGroups, smartDetectorAlertRules...}
Locations         : {global, North Central US, West Central US, East US...}
```

Step 2: Deploy zone-resilient Azure VM Scale Sets

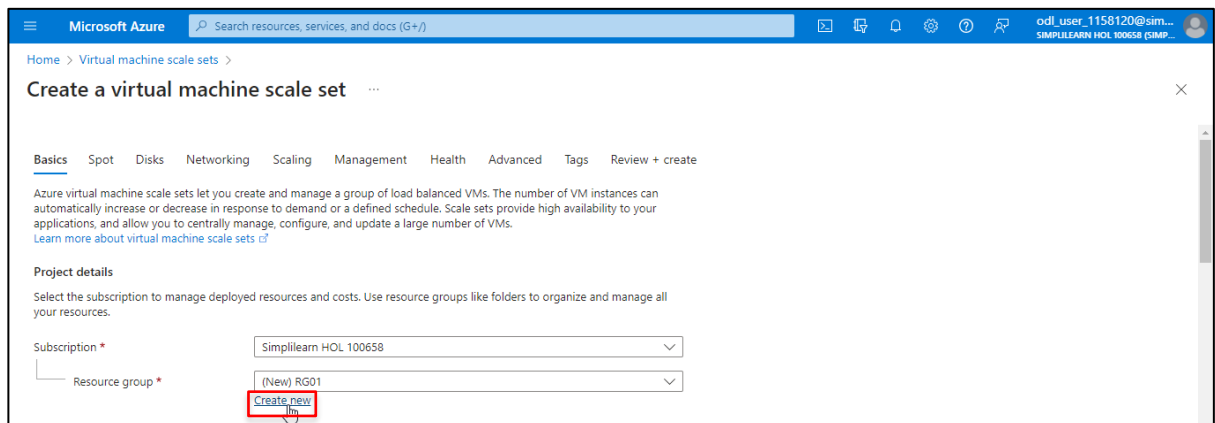
2.1 On the **Microsoft Azure** home page, search for and select **Virtual machine scale sets**



2.2 In the **Virtual machine scale sets** page, click on **+ Create**



2.3 On the **Create a virtual machine scale set** page, navigate to the **Basics** tab. Now, under **Project details**, click on **Create new** to create a new **Resource group RG01**.



2.4 Under **Scale set details**, provide the information as shown in the screenshot below:

Scale set details

Virtual machine scale set name *

Region *

Availability zone ⓘ

Autoscaling can help you respond to an outage by scaling out new instances in another zone. Turn on Autoscaling in the [Scaling tab](#).

Orchestration

A scale set has a "scale set model" that defines the attributes of virtual machine instances (size, number of data disks, etc). As the number of instances in the scale set changes, new instances are added based on the scale set model.
[Learn more about the scale set model](#)

Orchestration mode * ⓘ

☐ **Flexible:** achieve high availability at scale with identical or multiple virtual machine types
☒ **Uniform:** optimized for large scale stateless workloads with identical instances

Security type ⓘ

2.5 Under **Instance details**, provide the information as shown below:

Instance details

Image * ⓘ

[See all images](#) | [Configure VM generation](#)

This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ⓘ

☐ Arm64
☒ x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ
☐

Size * ⓘ

[See all sizes](#)

Enable Hibernation (preview) ⓘ
☐

To enable Hibernation, you must register your subscription. [Learn more](#)

2.6 Under **Administrator account**, provide the details as shown below:

Administrator account

Username * ⓘ

azureuser01 ✓

Password * ⓘ

..... ✓

Confirm password * ⓘ

..... ✓

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#)

Would you like to use an existing Windows Server license? ☐ ⓘ

[Review Azure hybrid benefit compliance](#)

2.7 Navigate to the **Disks** tab

Microsoft Azure

Search resources, services, and docs (G+)

odl_user_1158120@sim...
SIMPLILEARN HOL 100658 (SIMP...)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics

Spot

Disks

Networking

Scaling

Management

Health

Advanced

Tags

Review + create

Azure virtual machine scale sets let you create and manage a group of load balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule. Scale sets provide high availability to your applications, and allow you to centrally manage, configure, and update a large number of VMs.

[Learn more about virtual machine scale sets](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
Simplilearn HOL 100658

Resource group *
(New) RG01
[Create new](#)

Scale set details

Virtual machine scale set name *
myvmss ✓

Region *
(US) East US

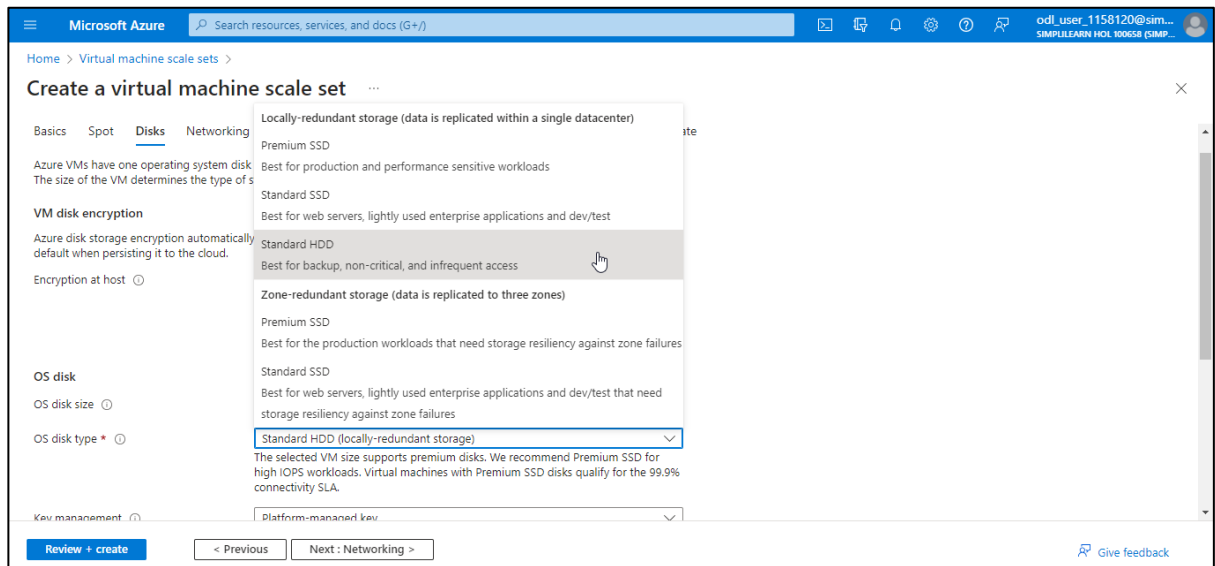
Review + create

< Previous

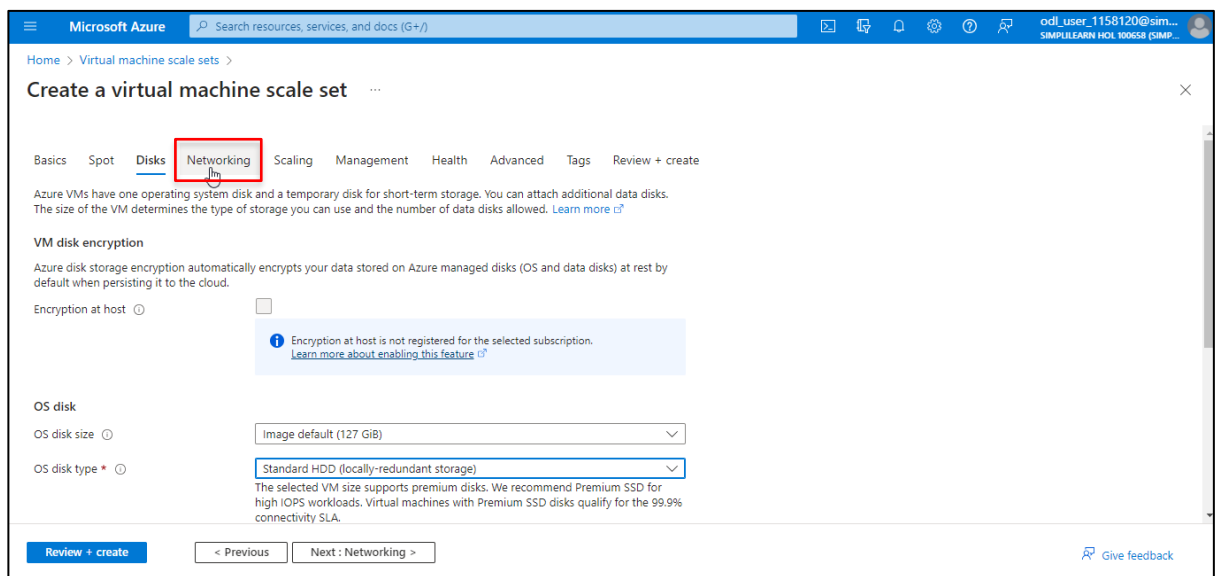
Next : Spot >

[Give feedback](#)

2.8 Select Standard HDD locally-redundant storage for OS disk type



2.9 Navigate to the Networking tab



2.10 Under **Virtual network configuration**, click on **Create virtual network**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks **Networking** Scaling Management Health Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more about VMSS networking](#)

Virtual network configuration

Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. [Learn more about VNets](#)

Virtual network * (New) RG01-vnet (recommended)
[create virtual network](#)

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

+ Create new nic Delete

☐ NAME CREATE PUBL... SUBNET NETWORK SECU... ACCELERATED NL...

[Review + create](#) < Previous Next: Scaling > [Give feedback](#)

2.11 In the **Create virtual network** page, provide the details as shown below and then click **OK**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks **Networking** Scaling Management Health

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more about VMSS networking](#)

Virtual network configuration

Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. [Learn more about VNets](#)

Virtual network * (New) RG01-vnet (recommended)
[Create virtual network](#)

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

+ Create new nic Delete

☐ NAME CREATE PUBL... SUBNET

Create virtual network

Virtual Network service enables Azure resources to securely communicate with each other in a virtual network which is a logical isolation of the Azure cloud dedicated to your subscription. You can connect virtual networks to other virtual networks, or your on-premises network. [Learn more about virtual networks](#)

Name * RG01-vnet ✓

Resource group * (New) RG01
[Create new](#)

Address space

The virtual network's address space specified as one or more address prefixes in CIDR notation (e.g. 10.0.0.0/16).

Address range	Addresses	Overlap
☒ 10.82.0.0/20 ✓	10.82.0.4 - 10.82.15.254 (4091 addresses)	None
☐ (0 Addresses)		None

Subnets

The subnet's address range in CIDR notation (e.g. 10.0.0.0/24). It must be contained by the address space of the virtual network.

Subnet name	Address range	Addresses
☒ subnet01 ✓	10.82.0.0/24 ✓	10.82.0.4 - 10.82.0.254 (251 addresses)

[Review + create](#) < Previous Next: Scaling > **OK** Discard

2.12 Under **Network interface**, click on **Edit network interface**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Virtual network * (New) RG01-vnet
[Create virtual network](#)

Network interface
A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

+ Create new nic Delete

NAME	CREATE PUBL...	SUBNET	NETWORK SECU...	ACCELERATED N...
RG01-vnet-nic01	No	subnet01 (10.82.0.0/24)	Basic	On

[Edit network interface](#)

Load balancing
You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options
☐ None
☒ Azure load balancer
 Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
☐ Application gateway

[Review + create](#) < Previous Next : Scaling > [Give feedback](#)

2.13 In the **Edit network interface** page, click on **Advanced** under **NIC network security group**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets > Create a virtual machine scale set >

Edit network interface

Network interface

Name * RG01-vnet-nic01

Virtual network (RG01-vnet)

Subnet * subnet01 (10.82.0.0/24)

NIC network security group
☐ None
☒ Basic
☐ Advanced

Public inbound ports *
☒ None
☐ Allow selected ports

[OK](#) [Cancel](#) [Give feedback](#)

2.14 Under **Configure network security group**, click on **Create new**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets > Create a virtual machine scale set >

Edit network interface

Network interface

Name *
RG01-vnet-nic01

Virtual network
RG01-vnet

Subnet *
subnet01 (10.82.0.0/24)

NIC network security group
☐ None
☐ Basic
☒ Advanced

Configure network security group *
 (new) myvmss-nsg
Create new

Public IP address
☒ Disabled ☐ Enabled

OK Cancel [Give feedback](#)

2.15 In the **Create network security group** page, click on **+ Add an inbound rule**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets > Create a virtual machine scale set > Edit network interface >

Create network security group

Name *
myvmss-nsg

Inbound rules

1000: default-allow-ssh	Any	SSH (TCP/22)	✓	
-------------------------	-----	--------------	---	--

+ Add an inbound rule

Outbound rules
No results.

+ Add an outbound rule

2.16 In the **Add inbound security rule** page, provide the details as shown in the screenshots below and click on **Add**:

Microsoft Azure

Home > Virtual machine scale sets > Create a virtual machine scale set > Edit network interface >

Create network security group

Name *

myvmss-nsg

Inbound rules

- 1000: default-allow-ssh
- Any
- SSH (TCP/22)
- + Add an inbound rule

Outbound rules

No results.

+ Add an outbound rule

OK

Add inbound security rule

myvmss-nsg

Source

Any

Source port ranges *

*

Destination

Any

Service

Custom

Destination port ranges *

80

Protocol

Any

☒ TCP

UDP

ICMP

Add

Cancel

Give feedback

Microsoft Azure

Home > Virtual machine scale sets > Create a virtual machine scale set > Edit network interface >

Create network security group

Name *

myvmss-nsg

Inbound rules

- 1000: default-allow-ssh
- Any
- SSH (TCP/22)
- + Add an inbound rule

Outbound rules

No results.

+ Add an outbound rule

OK

Add inbound security rule

myvmss-nsg

Protocol

Any

☒ TCP

UDP

ICMP

Action

☒ Allow

Deny

Priority *

1010

Name *

custom-allow-http

Description

Add

Cancel

Give feedback

2.17 Click OK

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets > Create a virtual machine scale set > Edit network interface >

Create network security group

Name *

myvmss-nsg

Inbound rules

1000: default-allow-ssh	Any	SSH (TCP/22)	✓	
1010: custom-allow-http	Any	HTTP (TCP/80)	✓	

+ Add an inbound rule

Outbound rules

No results.

+ Add an outbound rule

OK

2.18 Under Public IP address, select Enabled and click OK

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets > Create a virtual machine scale set > Edit network interface >

Edit network interface

Virtual network

RG01-vnet

Subnet *

subnet01 (10.82.0.0/24)

NIC network security group

☐ None
☐ Basic
☒ Advanced

Configure network security group *

(new) myvmss-nsg

Create new

Public IP address

Accelerated networking

OK Cancel

Give feedback

2.19 Under **Select a load balancer**, click on **Create a load balancer**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

RG01-vnet-nic01 Yes subnet01 (10.82.0.0/24) Advanced On

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options

- ☐ None
- ☒ **Azure load balancer**
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- ☐ Application gateway
Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

To allow traffic from your load balancing product, please update the appropriate port configuration on your network security group associated with your network interface.

Select a load balancer *

No existing load balancers in current subscription and location.

[Create a load balancer](#)

[Review + create](#) < Previous Next: Scaling > [Give feedback](#)

2.20 On the **Create a load balancer** page, provide the **Load balancer name** and click on **Create**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

RG01-vnet-nic01 Yes subnet01 (10.82.0.0/24) Advanced On

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options

- ☐ None
- ☒ **Azure load balancer**
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- ☐ Application gateway
Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

To allow traffic from your load balancing product, please update the appropriate port configuration on your network security group associated with your network interface.

Select a load balancer *

No existing load balancers in current subscription and location.

[Create a load balancer](#)

[Review + create](#) < Previous Next: Scaling >

Create a load balancer

Details such as subscription and resource group will be inherited from the virtual machine that you're creating. A default IP, backend pool, and load balancer rule will be created on your behalf, though certain configurations can be changed if desired.

Load balancer name *

vmss-lb

Type *

- ☒ **Public**
Provides outbound connections for virtual machines inside your virtual network using public load balancers.
- ☐ Internal
Used to load balance traffic inside a virtual network. A load balancer frontend can be accessed from an on-premises network in a hybrid scenario.

Protocol *

- ☒ **TCP**
- ☐ UDP

Rules

[Create](#) [Cancel](#)

2.21 Click on **Next : Scaling >**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks **Networking** Scaling Management Health Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more about VMSS networking](#)

Virtual network configuration

Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. [Learn more about VNets](#)

Virtual network * (New) RG01-vnet [Create virtual network](#)

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

+ Create new nic Delete

NAME	CREATE PUBL...	SUBNET	NETWORK SECU...	ACCELERATED N...
------	----------------	--------	-----------------	------------------

[Review + create](#) [< Previous](#) **Next : Scaling >** [Next : Scaling >](#) [Give feedback](#)

2.22 Provide **1** as the **Initial instance count**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks Networking **Scaling** Management Health Advanced Tags Review + create

An Azure virtual machine scale set can automatically increase or decrease the number of VM instances that run your application. This automated and elastic behavior reduces the management overhead to monitor and optimize the performance of your application. [Learn more about VMSS scaling](#)

Initial instance count * 1

Scaling

Scaling policy (Manual selected, Custom unselected)

Scale-In policy

Configure the order in which virtual machines are selected for deletion during a scale-in operation. [Learn more about scale-in policies](#)

Scale-in policy Default - Balance across availability zones and fault domains, then delete V...

Apply force delete to scale-in operations

[Review + create](#) [< Previous](#) [Next : Management >](#) [Give feedback](#)

2.23 Navigate to the **Management** tab

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks **Networking** Scaling **Management** Health Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more about VMSS networking](#)

Virtual network configuration

Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. [Learn more about VNets](#)

Virtual network * (New) RG01-vnet
[Create virtual network](#)

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

+ Create new nic Delete

NAME	CREATE PUBLI...	SUBNET	NETWORK SECU...	ACCELERATED NL...
------	-----------------	--------	-----------------	-------------------

Review + create < Previous Next : Scaling >

[Give feedback](#)

2.24 Under **Monitoring**, select **Enable with custom storage account**

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks Networking Scaling **Management** Health Advanced Tags Review + create

Configure monitoring and management options for your virtual machine scale set instances.

Microsoft Defender for Cloud

Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

✓ Your subscription is protected by Microsoft Defender for Cloud basic plan.

Upgrade policy

Upgrade mode * Manual - Existing instances must be manually upgraded

Monitoring

Boot diagnostics

☐ Enable with managed storage account (recommended)
☒ **Enable with custom storage account**
☐ Disable

Diagnostics storage account * (new) rg01diag387

Review + create < Previous Next : Health >

[Give feedback](#)

2.25 Click on **Review + create**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks Networking Scaling **Management** Health Advanced Tags Review + create

Configure monitoring and management options for your virtual machine scale set instances.

Microsoft Defender for Cloud
Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

✓ Your subscription is protected by Microsoft Defender for Cloud basic plan.

Upgrade policy
Upgrade mode ☐ Manual - Existing instances must be manually upgraded

Monitoring
Boot diagnostics ☐ Enable with managed storage account (recommended)
☒ Enable with custom storage account
☐ Disable

Review + create < Previous Next : Health > Give feedback

2.26 Once the validation is passed, click on **Create**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Virtual machine scale sets >

Create a virtual machine scale set

Basics Spot Disks Networking Scaling Management Health Advanced Tags **Review + create**

✓ Validation passed

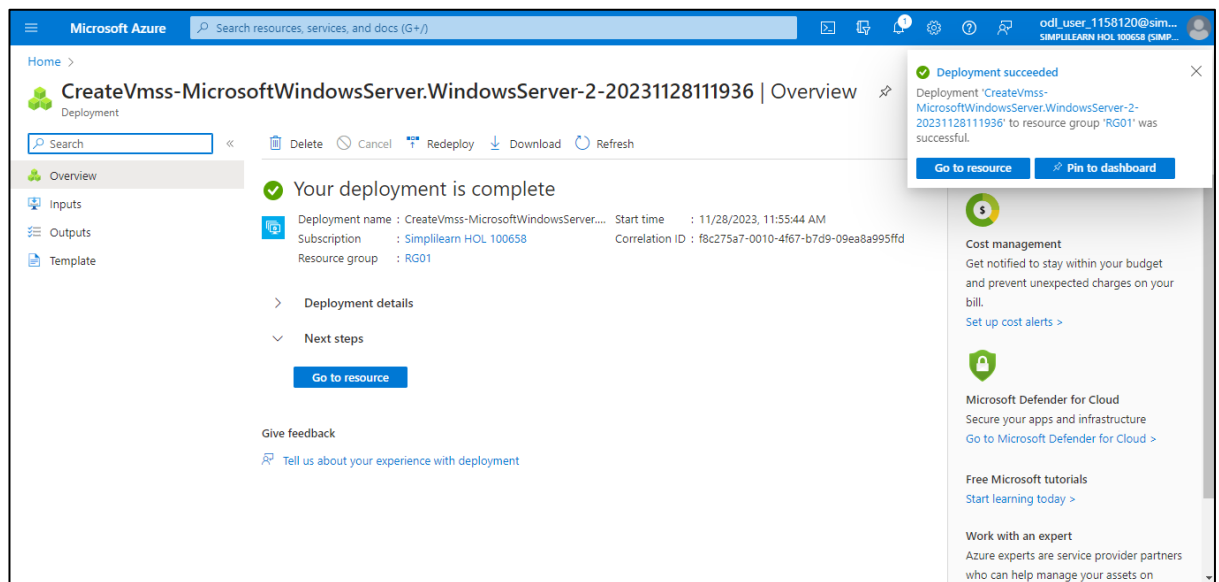
Basics

Subscription	Simplilearn HOL 100658
Resource group	(new) RG01
Virtual machine scale set name	myvmss
Region	East US
Orchestration mode	Uniform
Availability zone	1,2,3
Image	Windows Server 2019 Datacenter - Gen2
Size	Standard D2s v3 (2 vcpus, 8 GiB memory)
Security type	Standard
Username	azureuser01

Spot

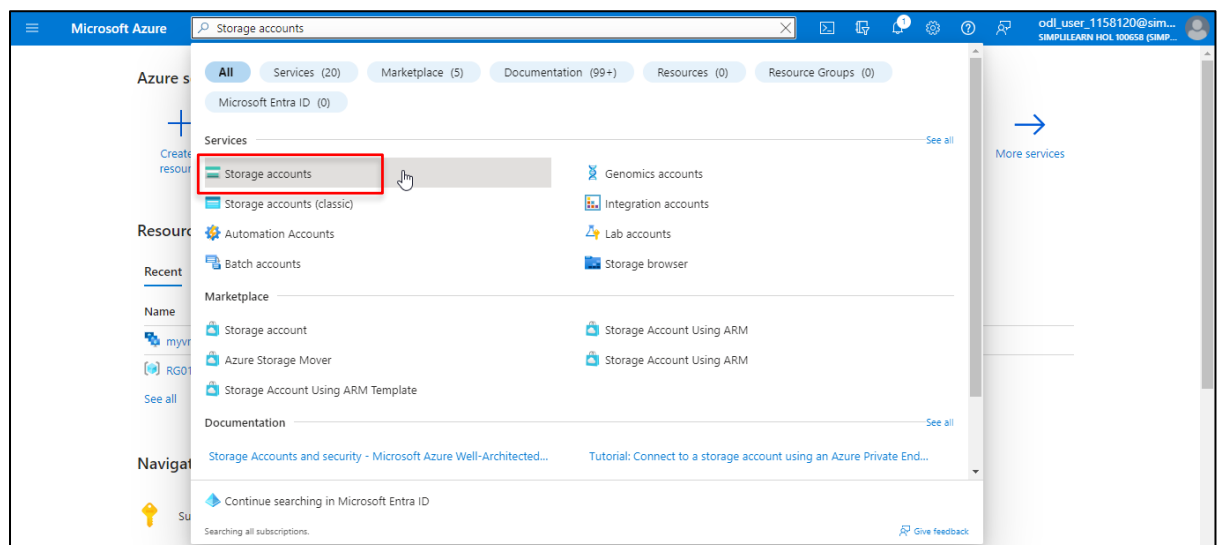
Create < Previous Next > Download a template for automation Give feedback

The deployment has been completed as shown in the screenshot below:

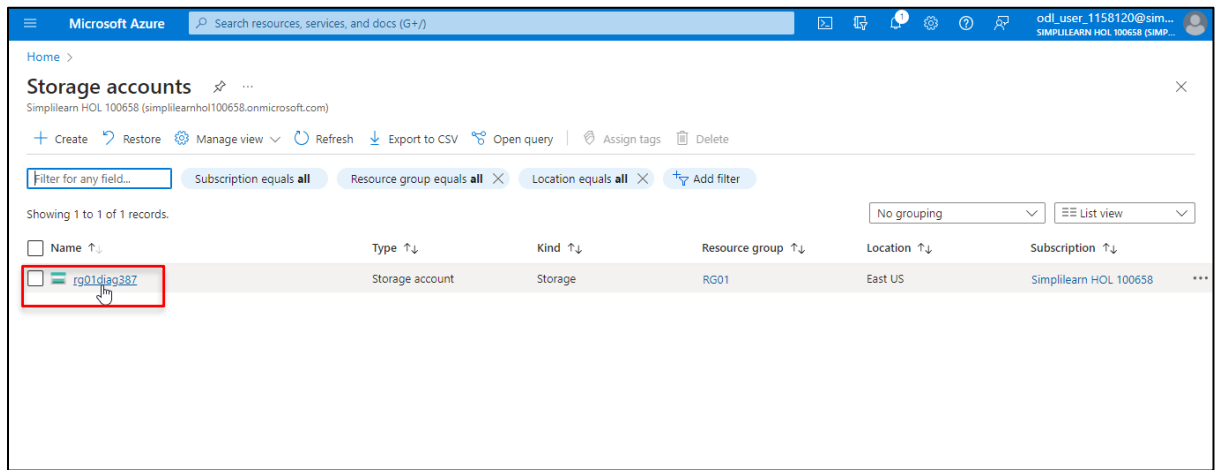


Step 3: Configure Azure VM Scale Sets using virtual machine extensions

3.1 In the Microsoft Azure home page, search for and select **Storage accounts**

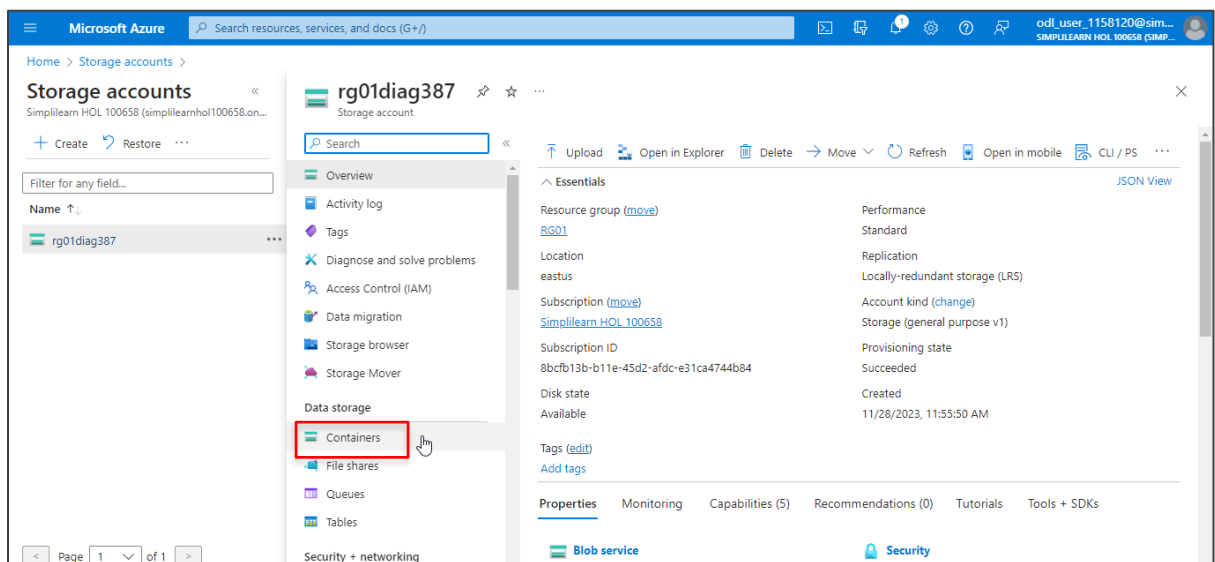


3.2 In the **Storage accounts** page, click on the storage account **rg01diag387**

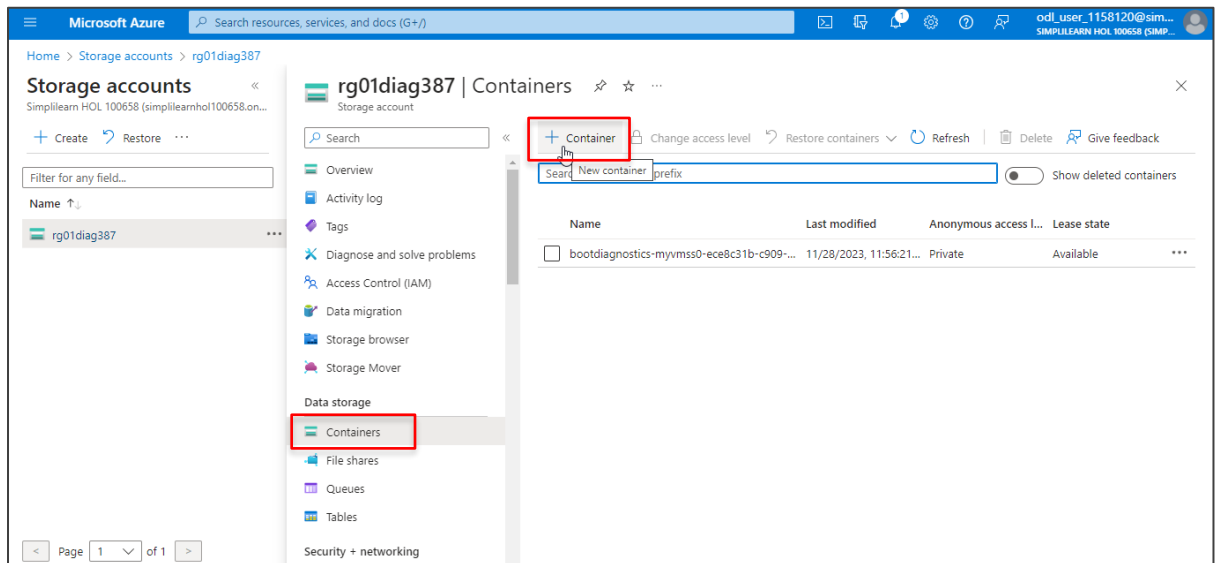


Note: The storage account is created automatically while creating VM Scale Sets.

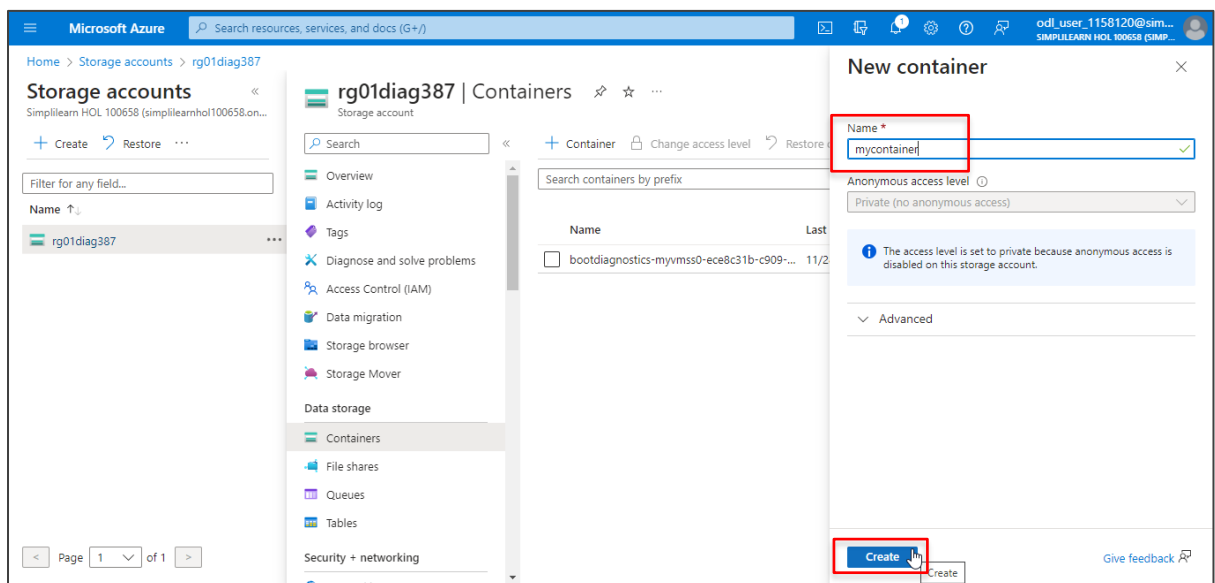
3.3 Click on **Containers** under **Data storage**



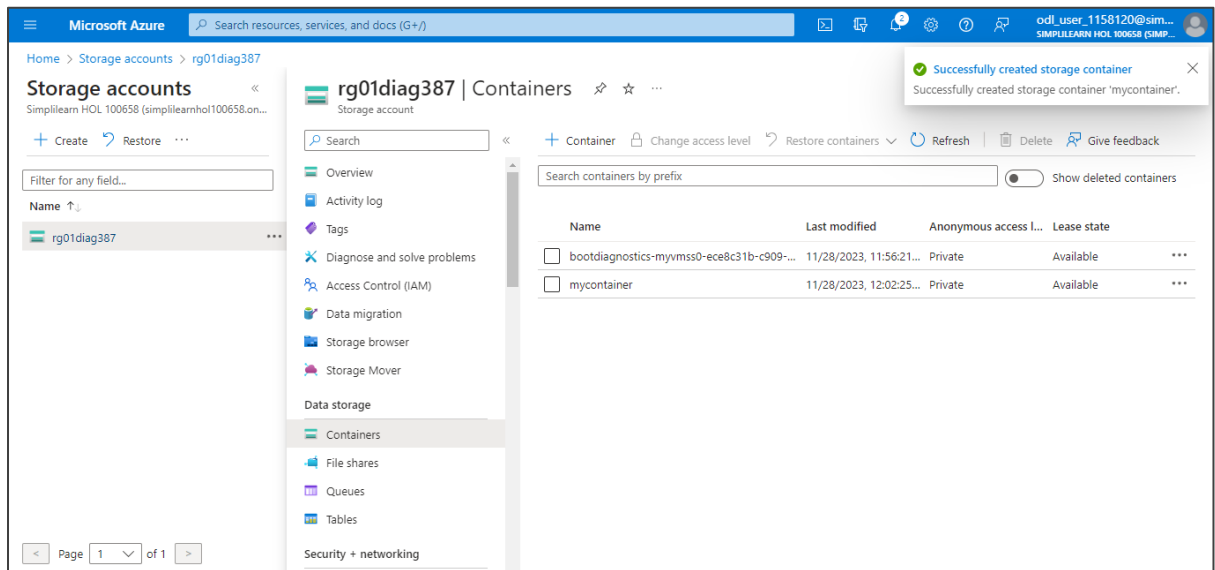
3.4 On the **Containers** page, click on **+ Container** to create a new container



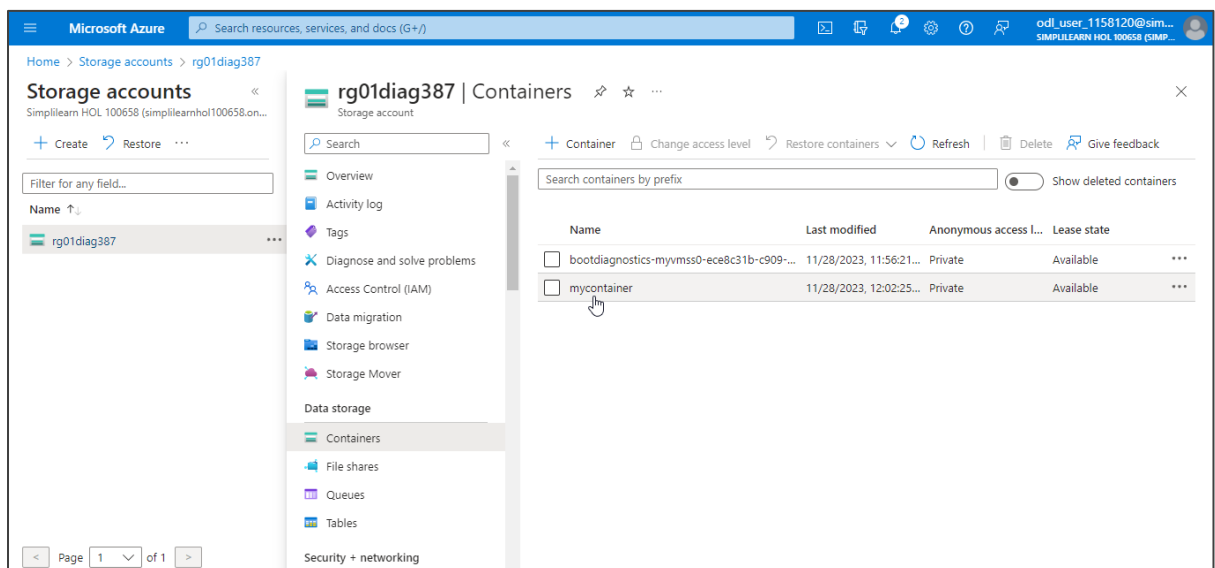
3.5 On the **New container** page, provide a **Name** and click **Create**



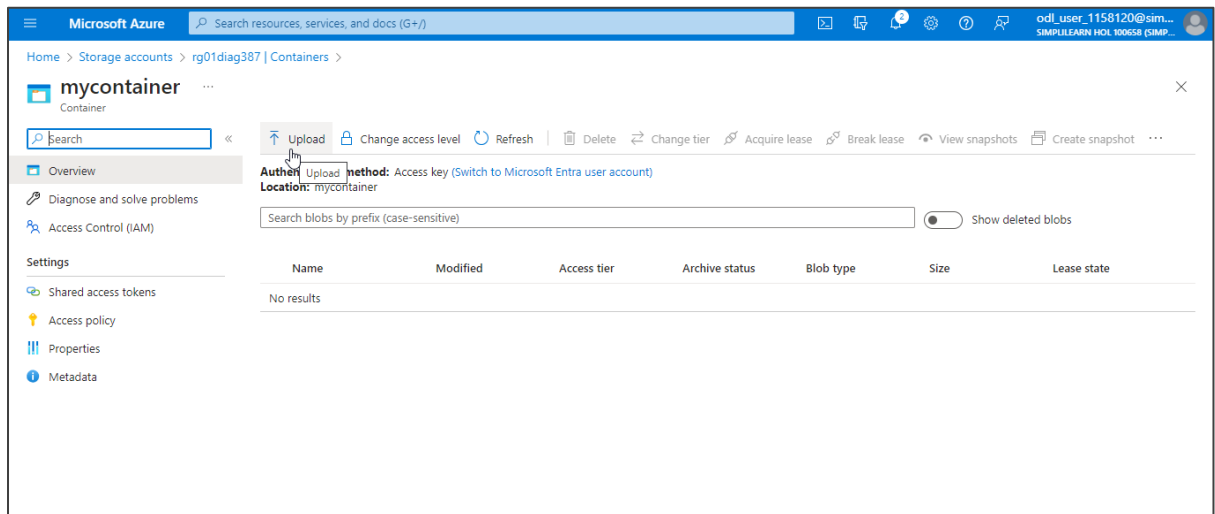
The container is created successfully as shown in the screenshot below:



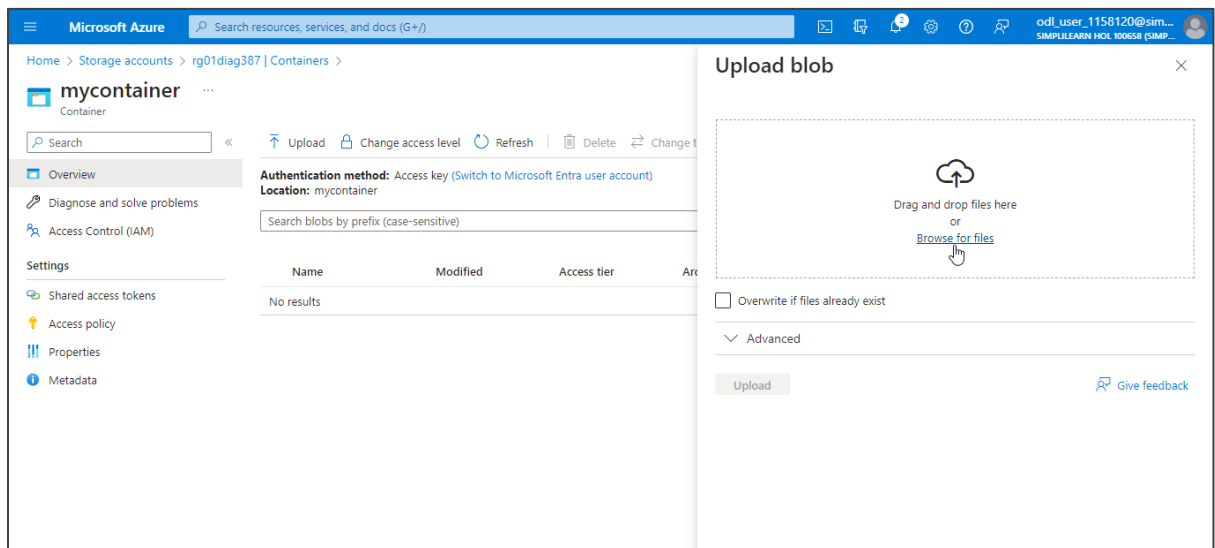
3.6 Click on the container (**mycontainer**) that was created



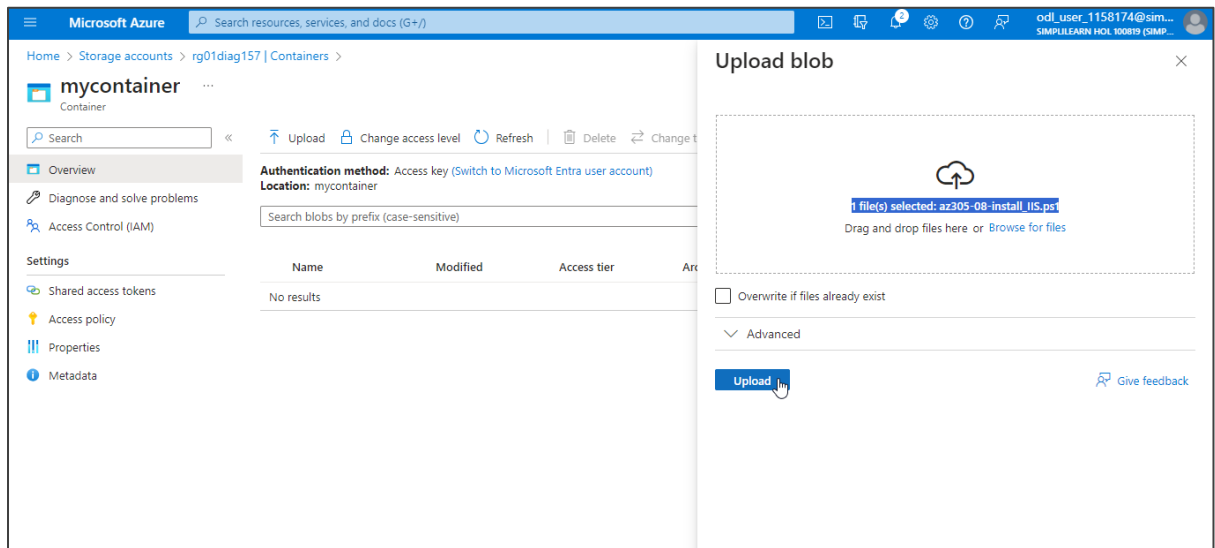
3.7 In the **mycontainer** page, click **Upload**



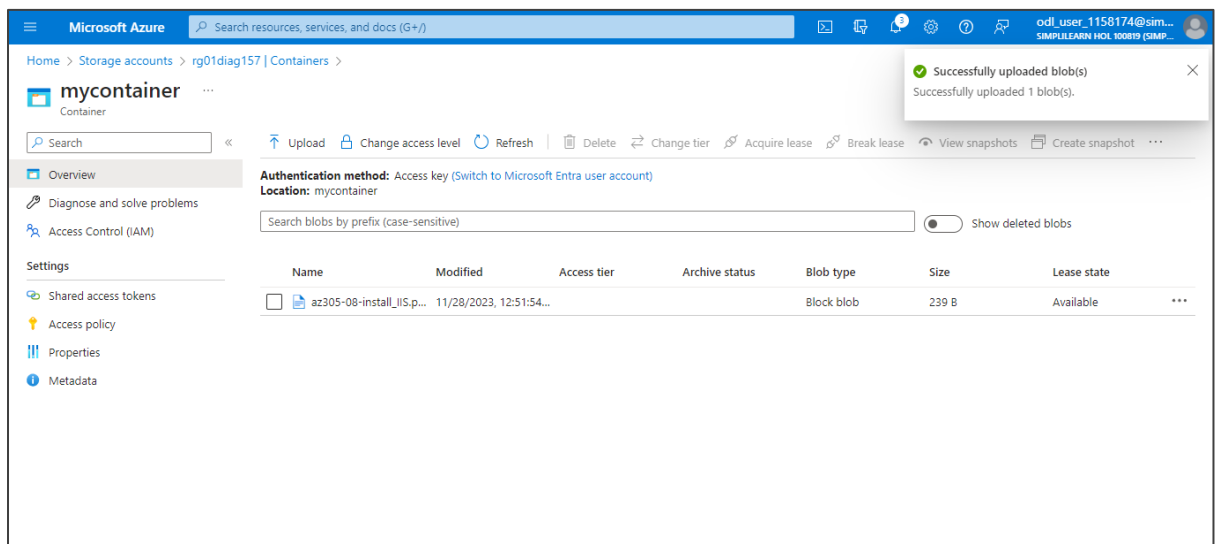
3.8 On the **Upload blob** page, click on **Browse for files** and upload **az305-08-install_IIS** file



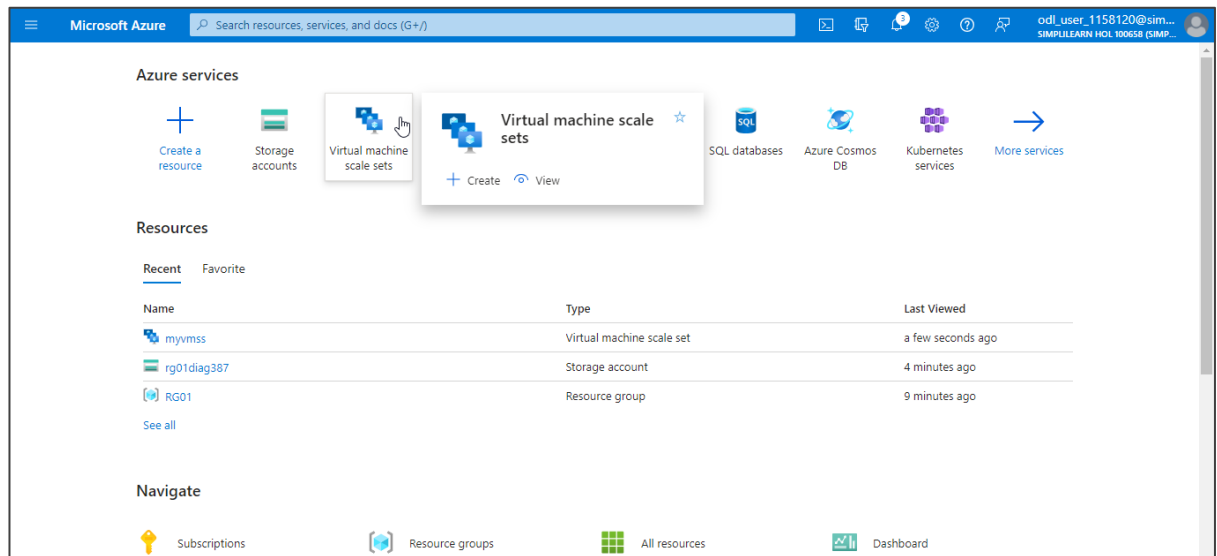
3.9 Once the file is selected, click **Upload**



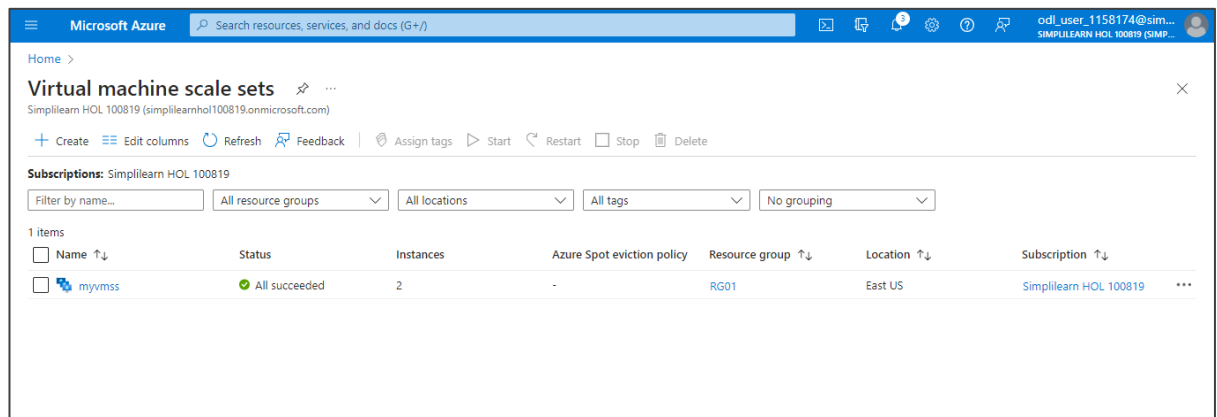
The blob is uploaded successfully as shown in the screenshot below:



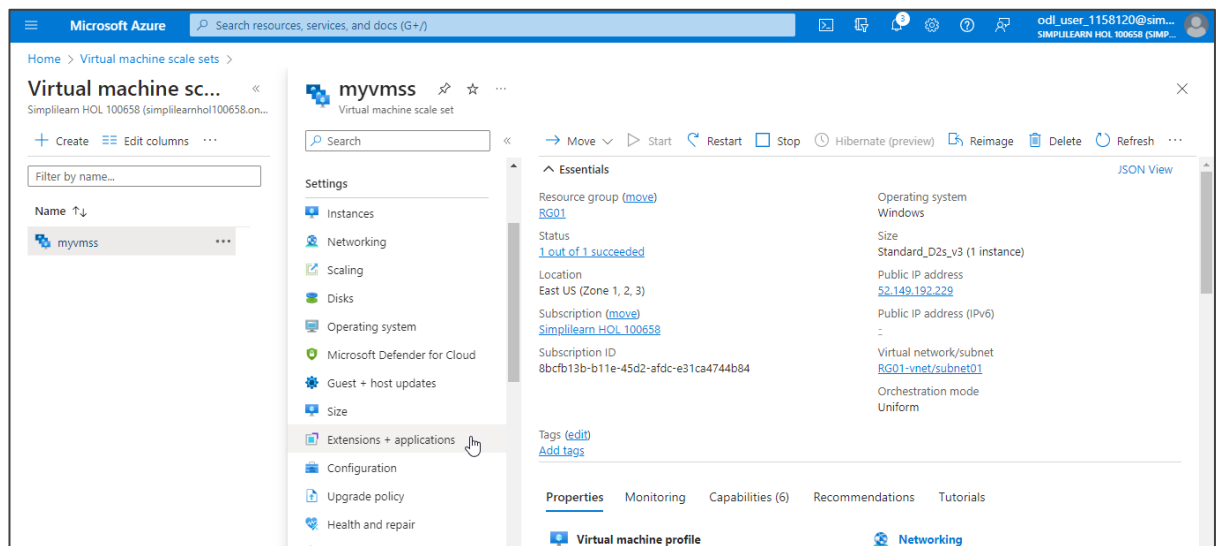
3.10 On the **Microsoft Azure** home page, click on **Virtual machine scale sets**



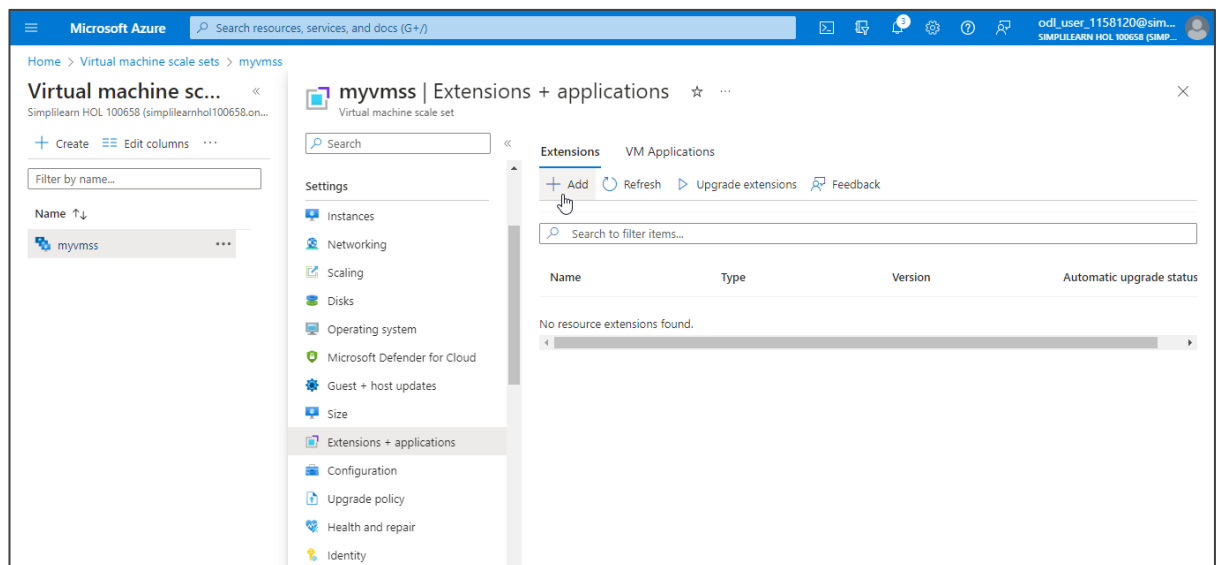
3.11 On the **Virtual machine scale sets** page, click on the VM Scale set (**myvmss**) that was created



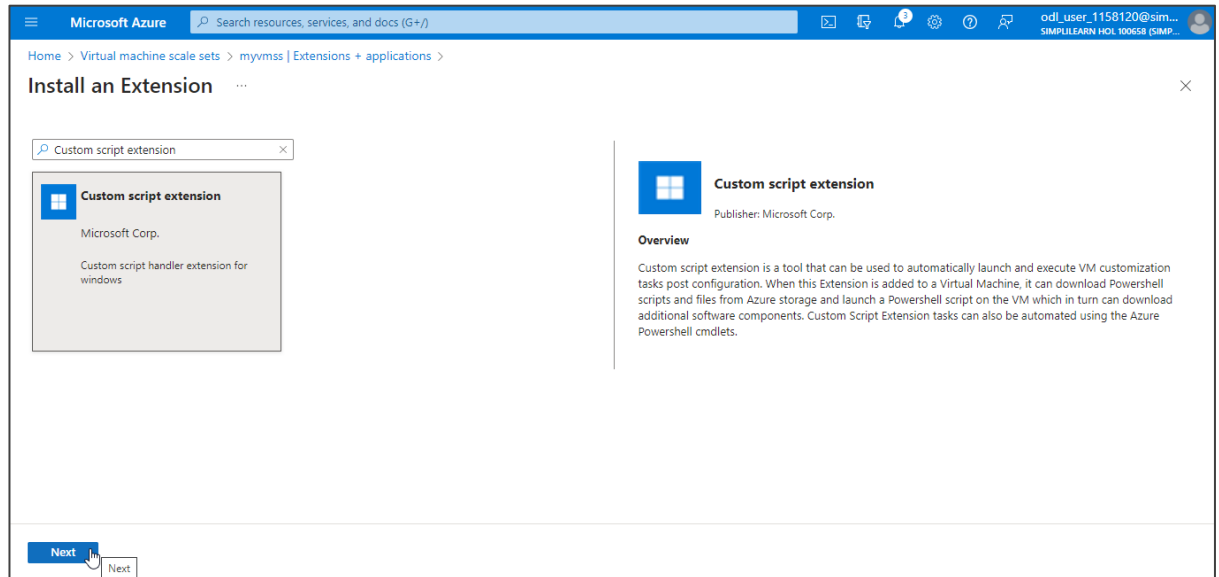
3.12 Click on **Extensions + applications** under **Settings**



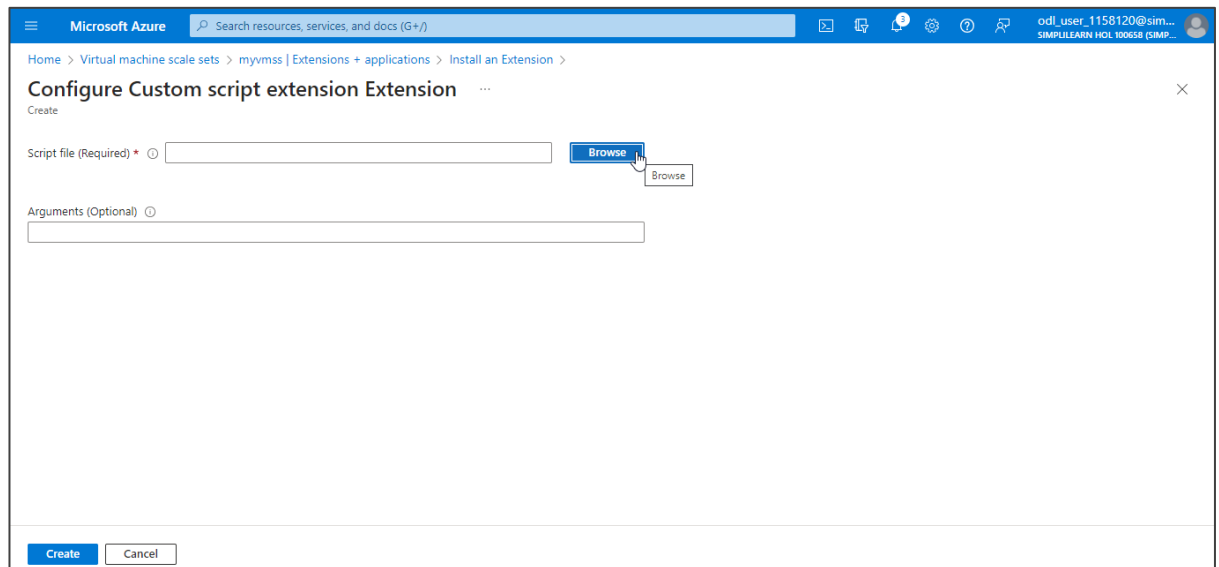
3.13 On the **Extensions + applications** page, click on **+ Add**



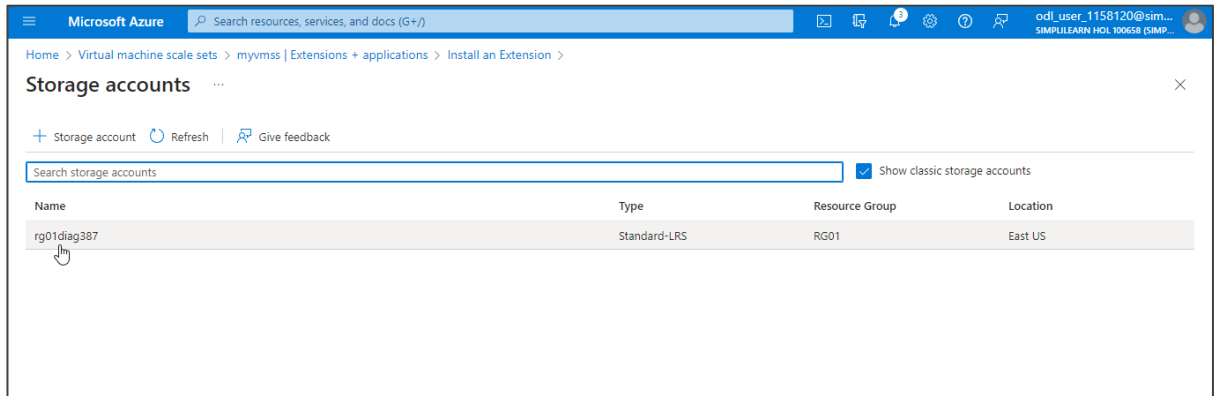
3.14 On the **Install an Extension** page, search for and select **Custom script extension** and click **Next**



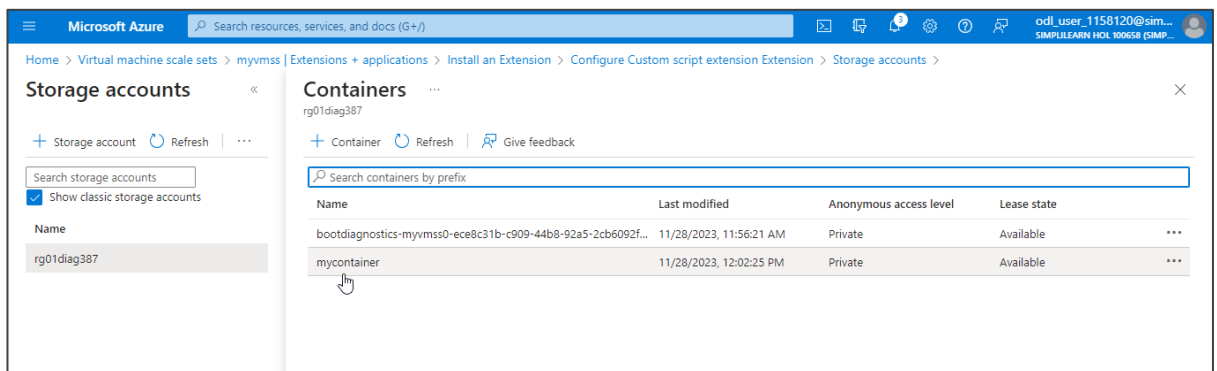
3.15 On the **Configure Custom script extension Extension** page, click **Browse**



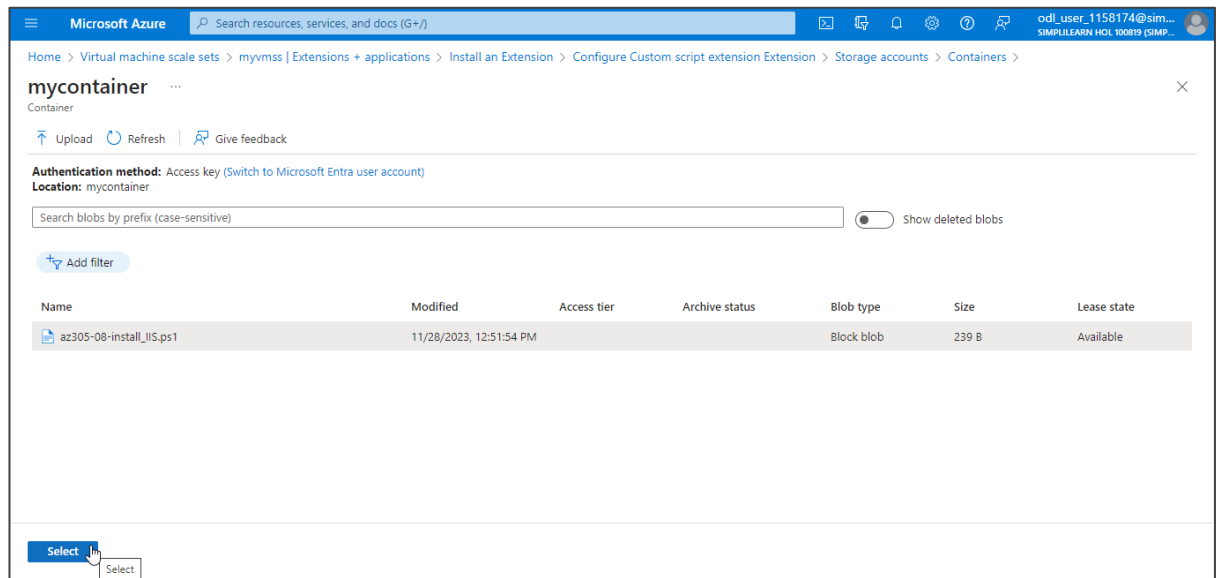
3.16 In the **Storage accounts** page, click on the previously created storage account (**rg01diag387**)



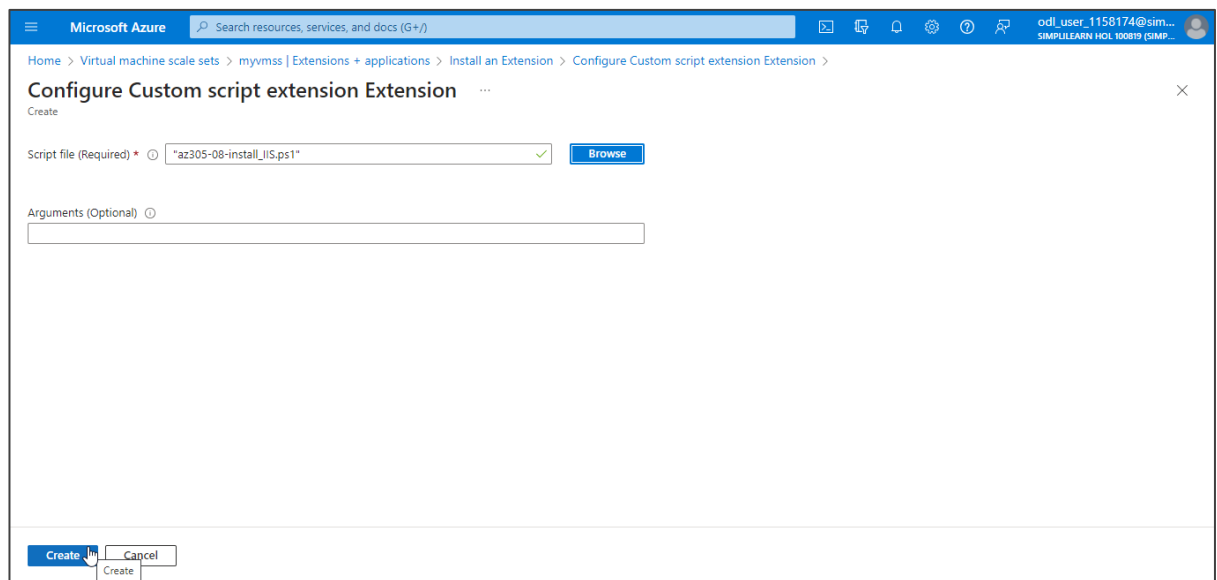
3.17 Click on the previously created container (**mycontainer**)



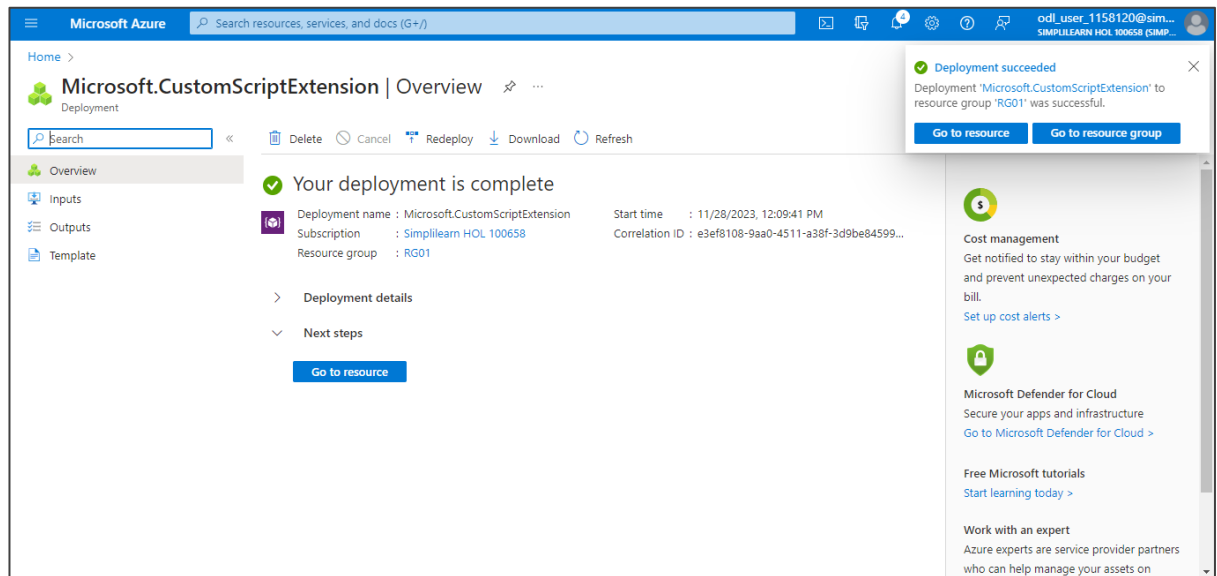
3.18 Select the blob file (az305-08-install_IIS.ps1) that was uploaded earlier and click **Select**



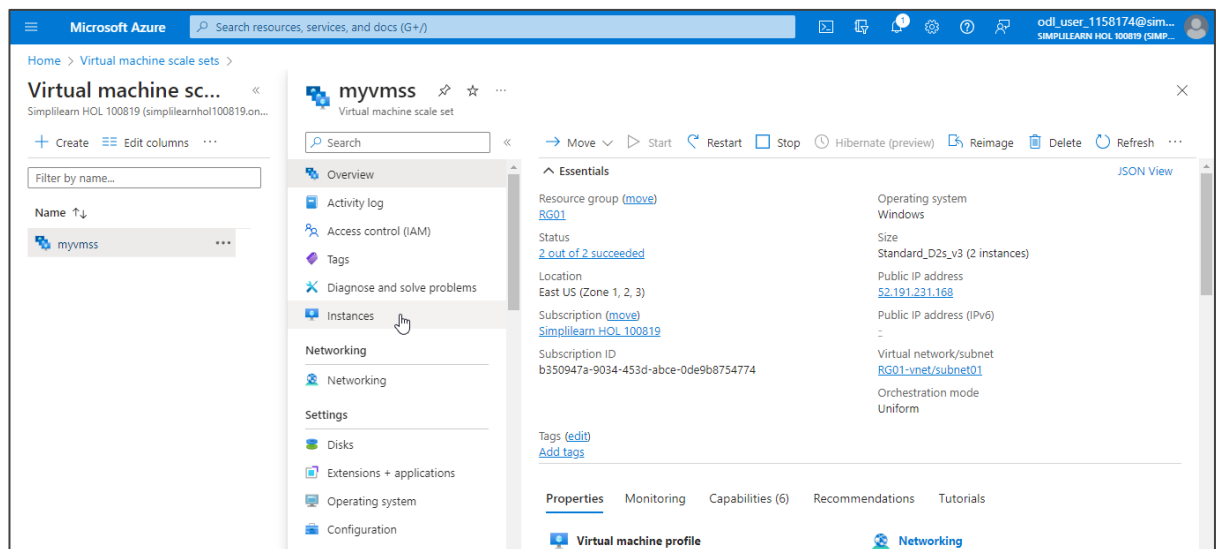
3.19 Click **Create**



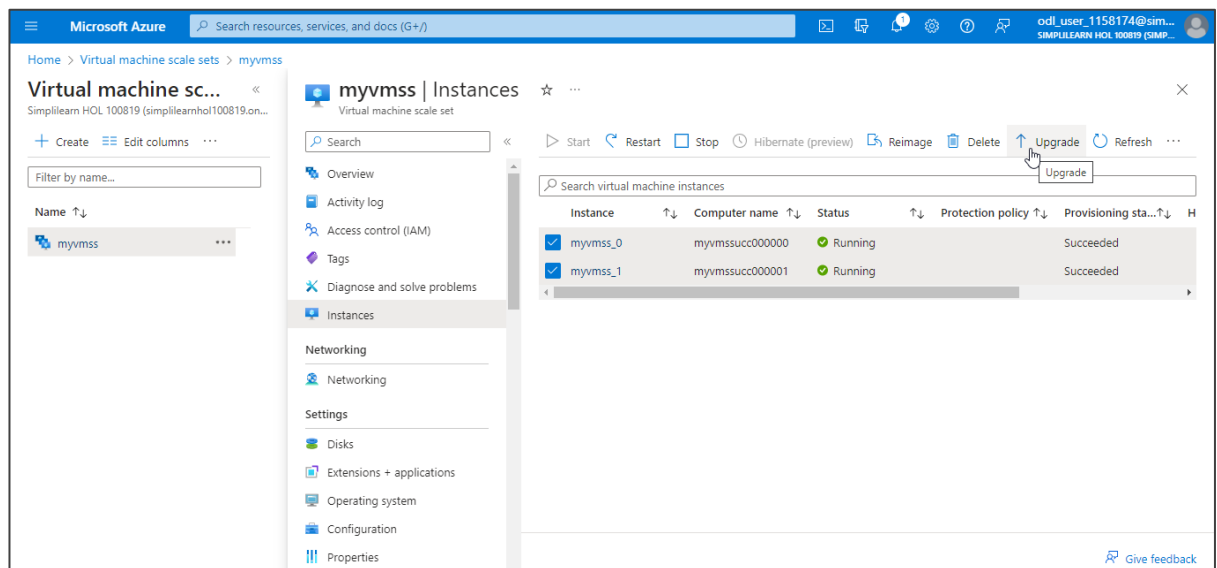
The deployment has been completed as shown in the screenshot below:



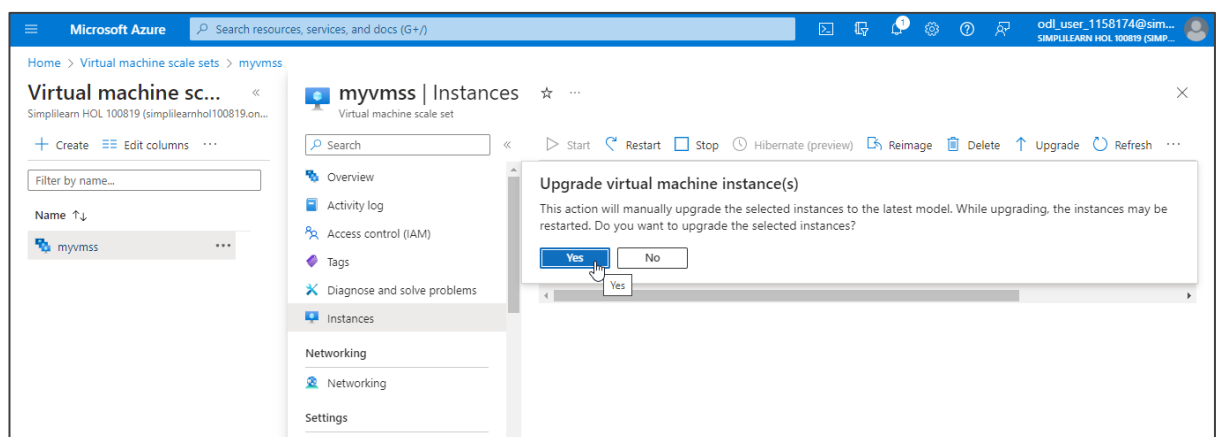
3.20 Navigate back to the VM Scale Set (myvmss) and click **Instances**



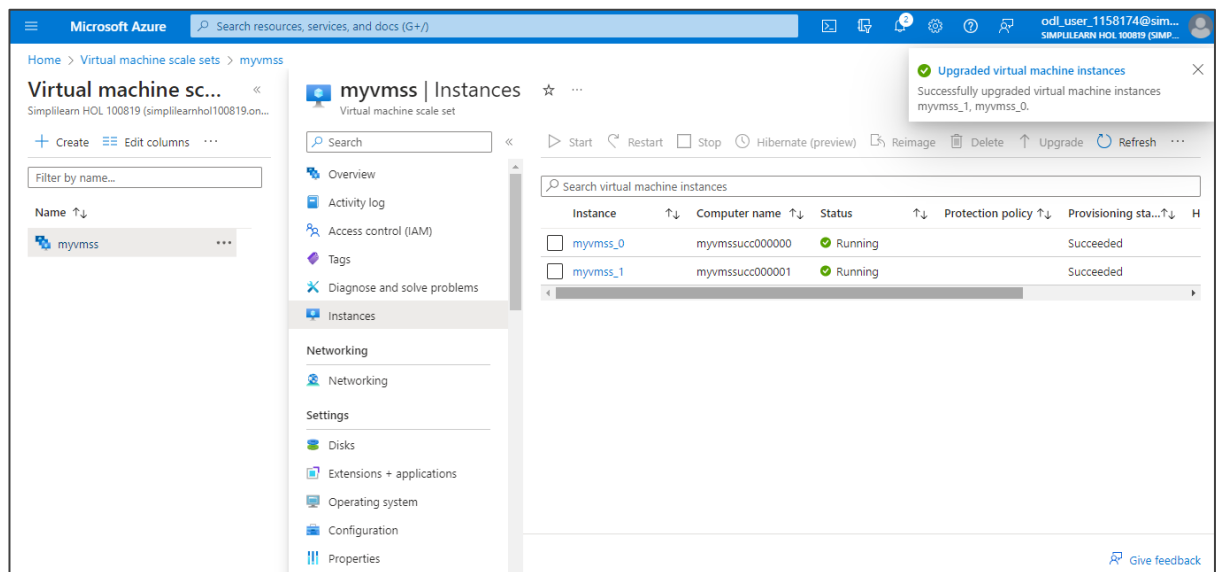
3.21 On the **Instances** page, check the two checkboxes and select **Upgrade**



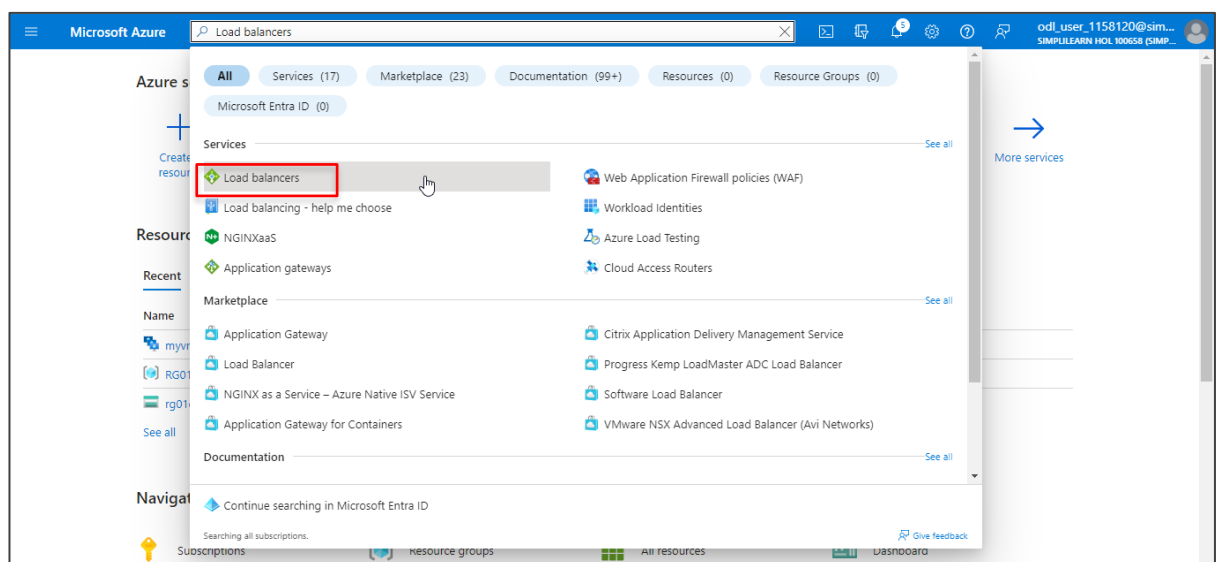
3.22 Click **Yes** when prompted



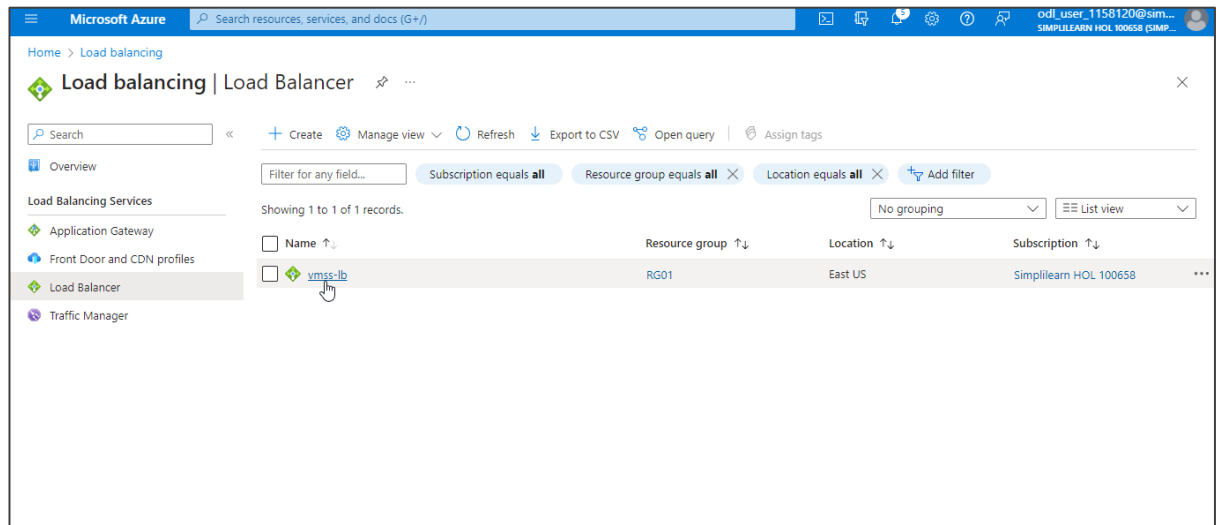
The virtual machine instances have been successfully upgraded as shown below:



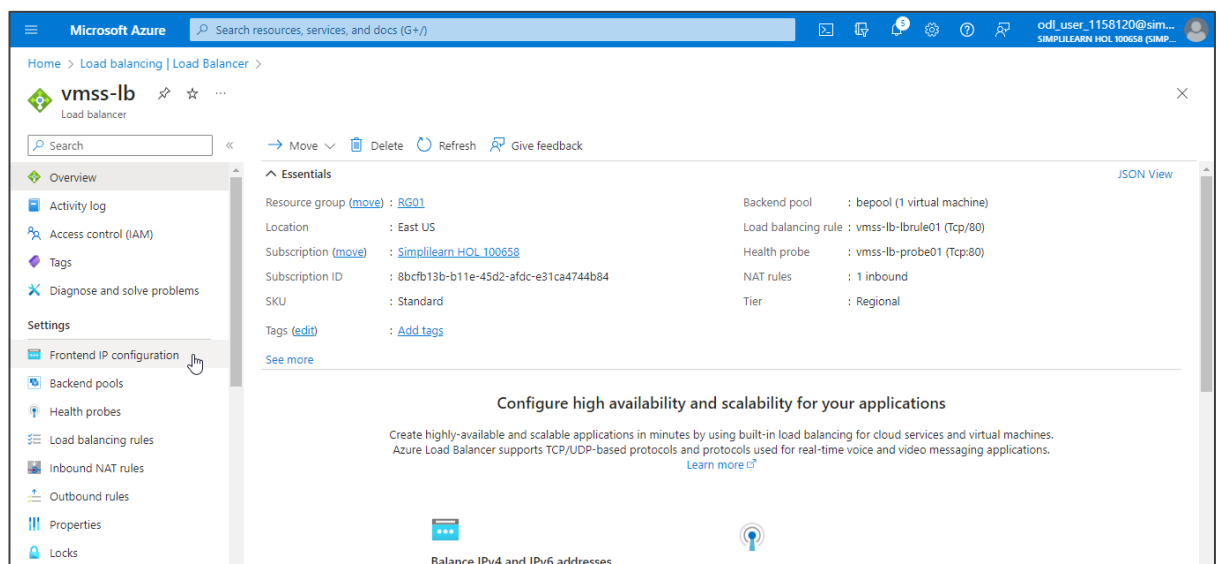
3.23 Navigate back to the **Microsoft Azure** home page, search for and select **Load balancers**



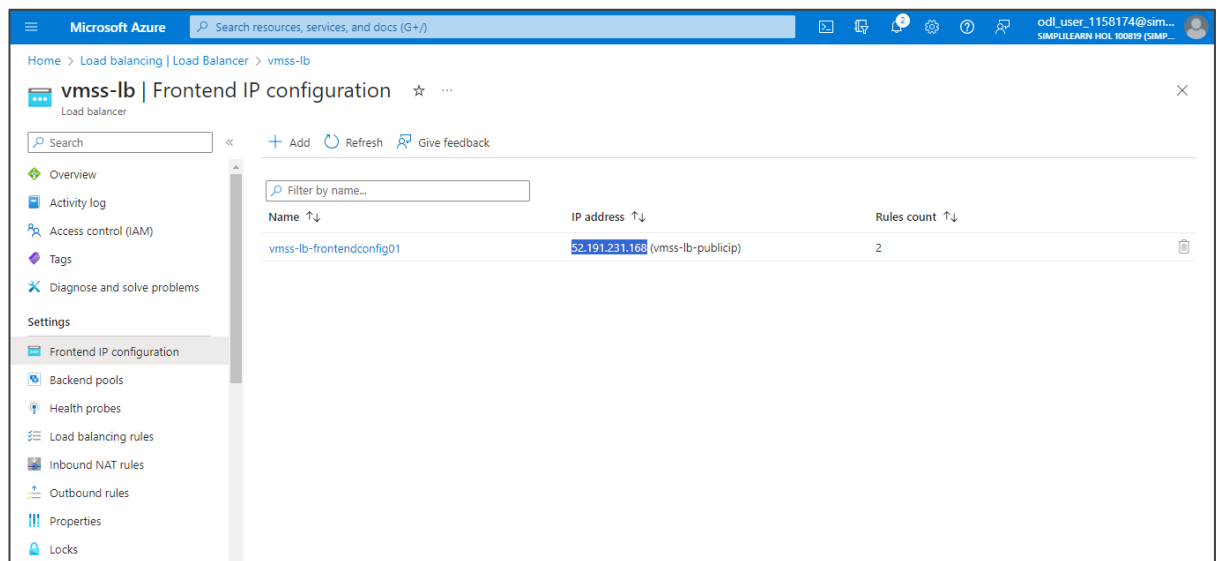
3.24 On the **Load Balancer** page, click on the load balancer **vmss-lb** that was created earlier



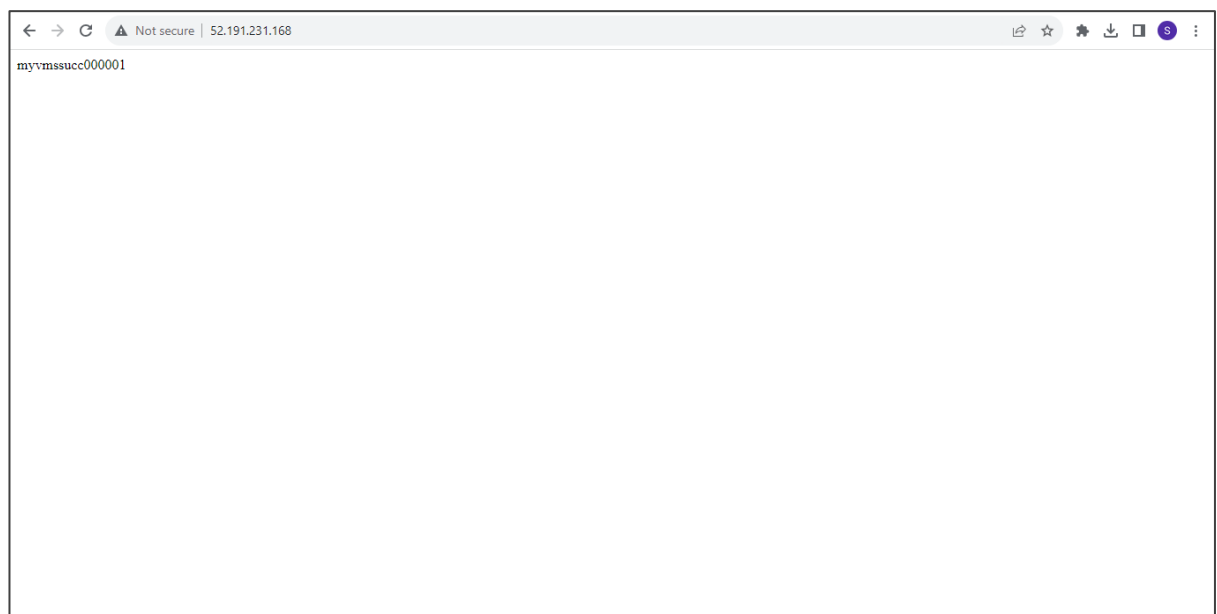
3.25 Click on **Frontend IP configuration** under **Settings**



Note the value of the IP address (**52.191.231.168**) as shown in the screenshot below:



3.26 Open a new browser tab and navigate to the IP address (**52.191.231.168**)



By following the above steps, you have successfully created virtual machines and VM Scale Sets using the Azure virtual machine Custom Script Extension.